

Test Yourself

Microeconomics

Changes in market

Signaling/Rationing

Indirect Tax (PED)

Subsidy (PED)

Dual changes in market

Section A

Duration of each question is **40 mins**

1. With the help of a diagram(s), explain what is meant by equilibrium in a market and consider the extent to which the equilibrium price and equilibrium quantity are likely to change for a product following an increase in the wages for labour across the whole economy. (8) **q3a/22/M/J/23**
2. With the help of a demand and supply diagram, explain how the introduction of an indirect tax affects equilibrium in a market and consider the extent to which the incidence of the tax will fall on the consumer. (8) **q2a/22/F/M/24**
3. Explain the functions of price in resource allocation and consider the importance of these functions in relation to the potential effectiveness of a market economy. (8) **q2a/22/F/M/23**

Section B

Duration **1 hour**

1. What would cause a market disequilibrium?
- A** when a decrease in the demand for a product leads to a price fall that results in some producers leaving the market
 - B** when a decrease in the demand for a product leads to some producers being unable to sell products at the current price
 - C** when a decrease in the supply of a product leads to a price rise that results in the market for the product clearing
 - D** when a decrease in the supply of a product leads to a price rise that makes it too expensive for some poorer households

2. In 2010, poor harvests reduced fruit crops and furniture producers gave big discounts on furniture products.

How might these changes have affected the position of the supply curve for the products?

	fruit	furniture producers
A	no change	shift to the left
B	shift to the left	no change
C	shift to the left	shift to the left
D	shift to the right	no change

3. A government wishes to raise the incomes of farmers without raising the price of food to consumers.

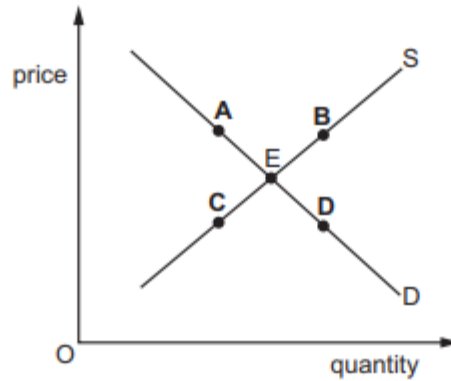
Which policy should it use?

- A** a maximum price below the market price for food
- B** a minimum price below the market price for food
- C** a payment of a subsidy to farmers to produce food
- D** a release of government food stocks onto the market

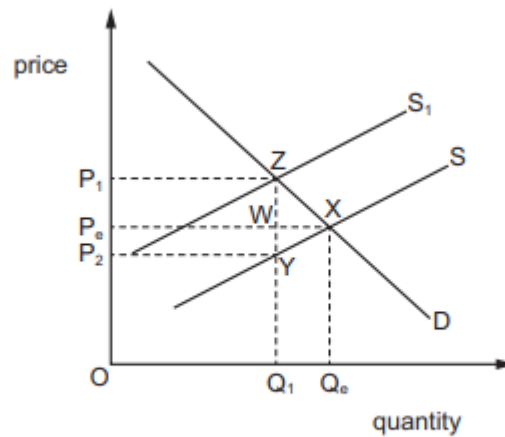
4. The diagram shows the demand for and supply of hotel accommodation. The market equilibrium is at point E.

The hotel receives a very large number of bad reviews about the quality of its accommodation.

Which point on the diagram would show the new market equilibrium?



5. The diagram shows the effect of an indirect tax imposed on cigarettes. The market is initially in equilibrium at point X.

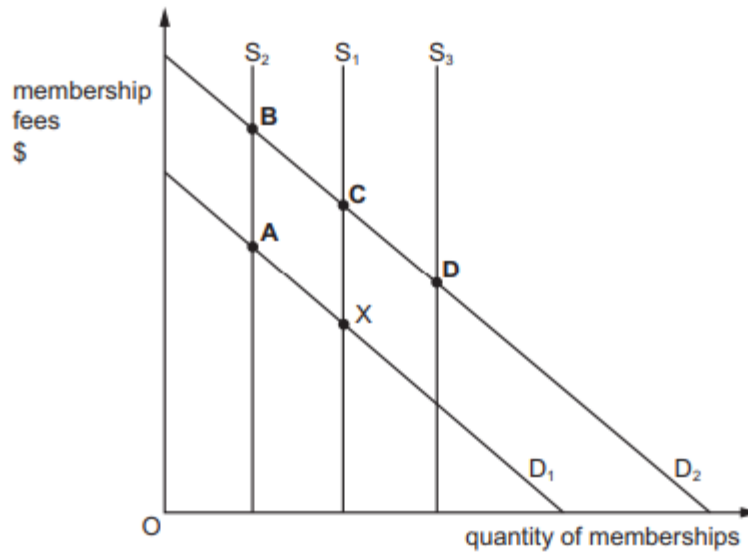


Which area represents the incidence of the tax on consumers?

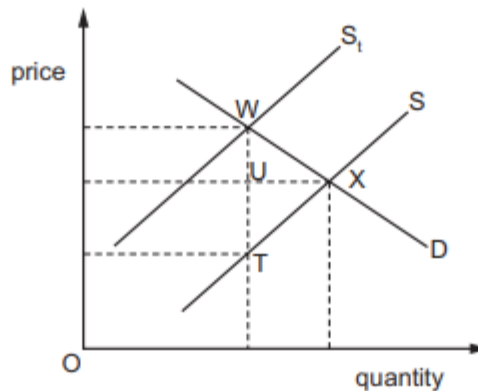
- A** P_1ZYP_2 **B** P_1ZWP_e **C** P_eWYP_2 **D** P_eXZP_1

6. Golf club membership fees in Singapore have risen as a result of restrictions on travel to cheaper golf courses in neighbouring countries and the closure of some local courses. The supply of golf club memberships is determined by the current number of courses.

If the original equilibrium was at X, which point shows the result of these changes?



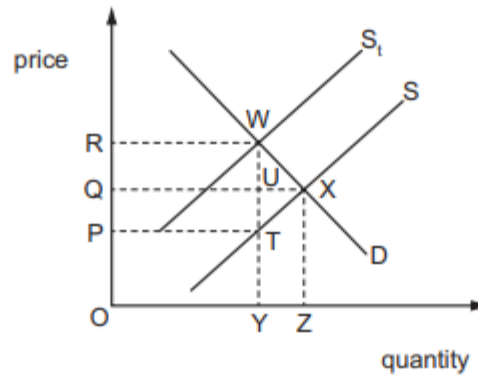
7. A specific tax is placed on the sale of bottles of lemonade. In the diagram, S is the supply curve before imposition of the tax and S_t is the supply curve after tax.



Which distance represents the specific tax on each bottle?

- A UT B WT C WU D WX

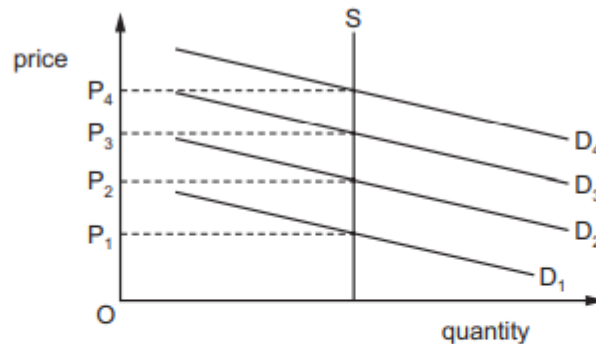
8. A specific tax is placed upon each bottle of perfume sold. In the diagram, S is the supply curve before tax, S_t is the supply curve after tax.



Which area represents that part of the tax revenue paid by producers?

- A** ORWY **B** PQUT **C** PRWT **D** QRWU

9. In the diagram, the supply curve shows the number of spaces in a car park and the demand curves show the demand for spaces on four different days, D_1 , D_2 , D_3 and D_4 .



The owner wishes to charge a parking fee on each of these days to allocate the spaces according to the market mechanism.

Which pricing policy should the owner use?

- A** set a fixed price at P_1
B set a fixed price at P_4
C vary prices between P_2 and P_3
D vary prices between P_1 and P_4

10. The pressure to reduce the use of fossil fuels has led to oil companies reducing the level of investment in the exploration of new reserves.

What will be the long-run effect of such a policy on the market for oil?

	market change	price change
A	upward movement along the demand curve	price decreases
B	downward movement along the demand curve	price increases
C	supply shifts to the left	price increases
D	supply shifts to the left	price decreases

11. When will the price mechanism **not** function as a system for allocating goods?

- A** when the government bans advertising
- B** when the government maintains an effective maximum price
- C** when there is a limited supply of the good
- D** when there is a powerful company able to set the market price

12. What does **not** happen when price acts as a means to allocate resources?

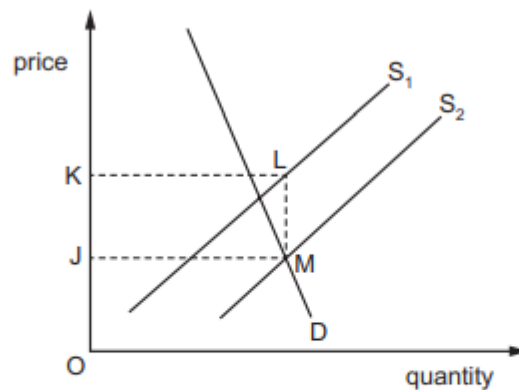
- A** Price determines the supply of public goods.
- B** Price operates in the markets for both goods and factors of production.
- C** Price recognises consumers' ability to pay rather than consumers' needs.
- D** Price signals to producers which goods are most profitable.

13. To improve the air quality for its citizens, a government introduces a subsidy for the producers of cars powered by batteries charged from the electricity supply.

What are the likely effects of this subsidy on the price and sales of electricity?

	price of electricity	sales of electricity
A	falls	falls
B	falls	rises
C	rises	falls
D	rises	rises

14. What is necessary to achieve the best allocation of resources using the price mechanism?
- A Producers can maximise their collective interest.
 - B Consumers and producers must have perfect knowledge.
 - C The government must be able to apply maximum and minimum prices.
 - D The value of the price elasticity of supply must be greater than one.
15. The diagram shows the effect on the market for rice of a change in government policy that causes a shift in the supply curve from S_1 to S_2 .

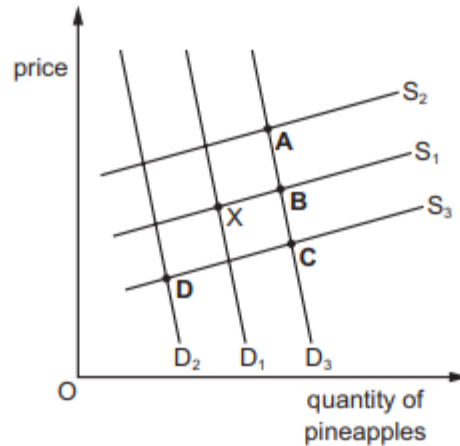


What does the area JKLM represent?

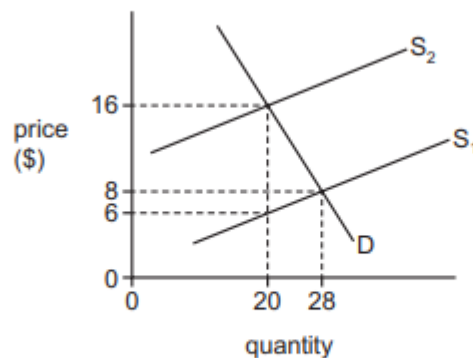
- A the cost to the government of a subsidy to rice growers
 - B the extra saving to importers of the removal of a tariff on rice
 - C the increase in consumer surplus from the introduction of a maximum price for rice
 - D the loss in government revenue from the reduction in a lump sum tax on rice
16. What is **not** a producer subsidy?
- A a grant given to cover losses in an essential industry, such as the rail industry
 - B a guaranteed minimum price paid to farmers to ensure the long run supply of food
 - C financial assistance given to a pharmaceutical company to develop a new vaccine
 - D a payment made to the unemployed to allow them to buy essential goods and services

17. The diagram shows the market for pineapples. X is the original equilibrium point. Pineapple producers benefit from improved technology. At the same time, a new diet book suggests that eating pineapples helps weight loss.

Which point is the new equilibrium?



18. The diagram shows demand and supply curves for a good. S_1 is the original supply curve, S_2 is supply after a tax is added.

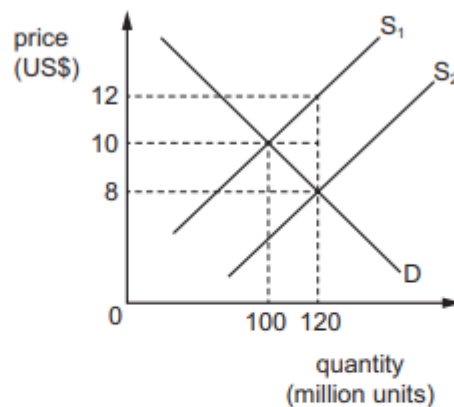


How much tax revenue is earned by the government?

- A** \$40 **B** \$160 **C** \$200 **D** \$320
19. The market for a product is initially in equilibrium.
- Which combination of changes will cause the price of the product to rise and the quantity sold to fall?
- A** a decrease in demand that is larger than a decrease in supply
B a decrease in demand that is smaller than a decrease in supply
C an increase in demand that is larger than an increase in supply
D an increase in demand that is smaller than an increase in supply

20. What does the incidence of an indirect tax on a product refer to?
- A the proportion of the tax paid by the consumer relative to the producer
 - B the reduction in output as a result of the imposition of the tax
 - C the revenue the tax generates for the government
 - D the increase in the hidden economy as a result of tax evasion after the tax is imposed

21. The diagram shows the effect of the government introducing a subsidy on a product.

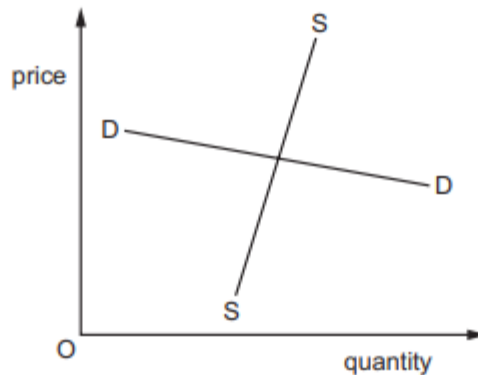


What are the effects of the subsidy on government spending and consumer spending?

	government spending	consumer spending
A	increases by US\$240 million	decreases by US\$40 million
B	increases by US\$240 million	decreases by US\$200 million
C	increases by US\$480 million	decreases by US\$40 million
D	increases by US\$480 million	decreases by US\$200 million

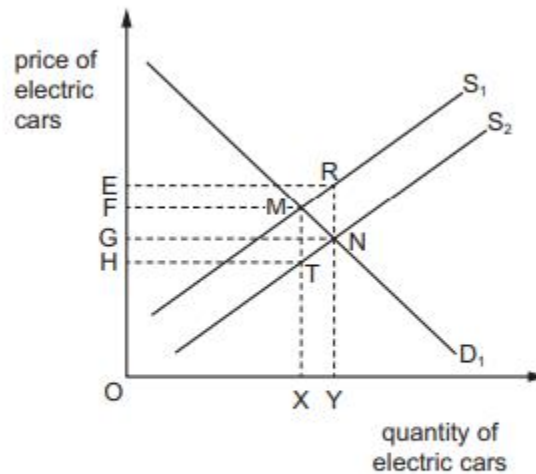
22. What is an example of the rationing function of the price mechanism?
- A When the price of a product is falling, firms will allocate less resources to its production.
 - B When the price of a product is falling, the government can set an effective minimum price.
 - C When the price of a product is rising, firms will expand production of the product to make profits.
 - D When the price of a product is rising, some consumers will no longer be able to buy it.

23. The diagram shows the demand curve and supply curve for a good on which the government imposes a specific tax.



What will be the result of this tax?

- A Most of the incidence of the tax will fall on the producer.
 - B There will be a new demand curve parallel to DD.
 - C The price will rise by the full amount of the tax.
 - D The quantity bought will fall proportionately to the tax rate.
24. The diagram represents the original demand curve D_1 and the original supply curve S_1 for electric cars. The government decides to provide a subsidy to electric car manufacturers.



Which area represents the total revenue of the car manufacturer after receiving the subsidy from the government?

- A OERY
- B OFMX
- C OGNV
- D OHTX

25. The table shows the market demand and market supply for kiwifruit over a year.

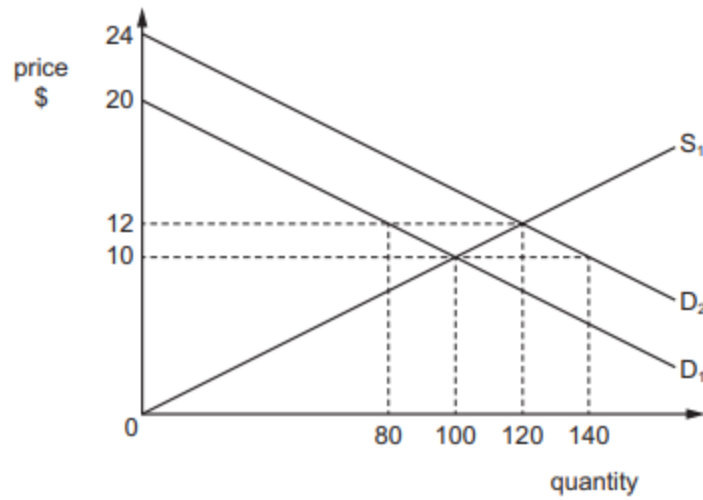
At a market price of \$3 per kg there is disequilibrium in the market.

price of kiwifruit (\$ per kg)	total market demand (thousand tonnes)	total market supply (thousand tonnes)
1	220	120
2	190	130
3	160	140
4	130	150
5	100	160

Which action would the government have to take to achieve market equilibrium at a price of \$3 per kg?

- A impose an indirect tax of \$2 per kg on kiwifruit
 - B purchase the entire supply at \$3 per kg and sell at \$2 per kg
 - C set a maximum price of \$3 per kg
 - D subsidise kiwifruit production by \$2 per kg
26. What is a major function of the price mechanism?
- A providing incentive for government intervention to reduce income inequality
 - B removing shortages by creating incentives for market prices to fall
 - C removing surpluses by creating incentives for market prices to rise
 - D signalling changes in market conditions to producers and consumers
27. In the market for a good the quantity supplied (Q_S) and the quantity demanded (Q_D) are given by $Q_S = P - 30$ and $Q_D = 240 - 2P$ where P = price in dollars.
- A change in the tax on the good makes $Q_S = P - 36$.
- How will the change affect equilibrium price?
- A It will fall by \$2.
 - B It will fall by \$6.
 - C It will rise by \$2.
 - D It will rise by \$6.

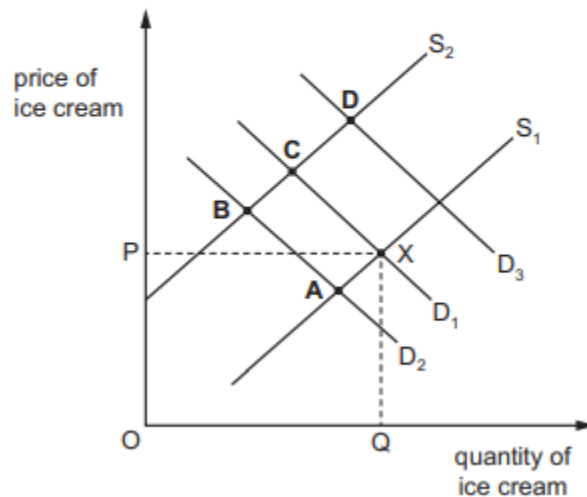
28. The diagram shows the market for a good with an initial equilibrium price of \$10. The demand for the good increases by 40 units at every price causing the equilibrium price to rise to \$12.



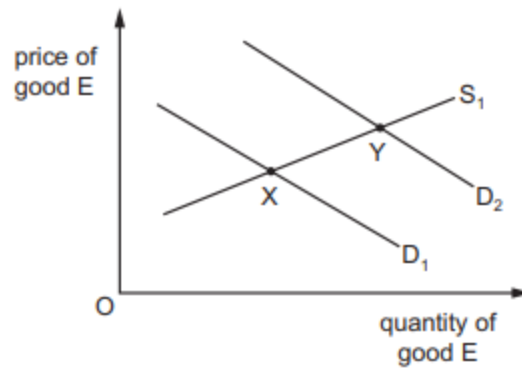
What is the value of the producer surplus after the price increase?

- A \$500 B \$720 C \$1000 D \$1440
29. Vanilla is an important ingredient in the production of ice cream. Between 2011 and 2018 the price of vanilla increased from about US\$100 per kg to nearly US\$600 per kg. The diagram shows the market demand for and supply of ice cream. The original equilibrium is X.

Which point illustrates the effect of this price rise on the market for ice cream?



30. The diagram shows the market demand for and supply of good E.



The equilibrium has changed from X to Y.

Which statement is **not** correct?

- A** Incomes of consumers may have increased and good E is a normal good.
- B** The quantity supplied has risen because price has increased.
- C** The quantity supplied has risen because the production of good E has been subsidised.
- D** The price of good F, which is a substitute for good E, may have risen.

Question	Answer	Marks
1	With the help of a diagram(s), explain what is meant by equilibrium in a market, and consider the extent to which the equilibrium price and equilibrium quantity are likely to change for a product following an increase in the wages for labour across the whole economy. Follow the point-based marking guidance at the top of this mark scheme and award: up to 3 marks for AO1 Knowledge and understanding up to 3 marks for AO2 Analysis up to 2 marks for AO3 Evaluation. AO1 Knowledge and Understanding (max 3 marks) The equilibrium point is where the quantity demanded equals the quantity supplied (1) and there is no tendency to change in a market (1) with an accurately labelled diagram showing the equilibrium price and equilibrium quantity (1) AO2 Analysis (max 3 marks) Note: there are two possible approaches to answering this question. If both the approach focused on demand-side (inferior and normal goods) and the supply-side approach are attempted, then only the best response should be marked including marks for evaluation. EITHER: The demand curve will be expected to shift in this situation. Whether it shifts to the left or to the right will depend on the nature of the product and will shift to the left if it is an inferior good or to the right if it is a normal good. This should be the basis of the discussion. For an accurately labelled diagram demonstrating at least one of the appropriate shift(s) in the demand curve i.e., to the right and / or the left (1) that clearly shows the change in the equilibrium price and the equilibrium quantity for both a normal and inferior good (1) and is accompanied by a clear explanation of both shifts (1) If only one shift (normal or inferior goods) is explained, then a maximum of 2 marks. Note: it is acceptable to use the same diagram for both knowledge and understanding and analysis OR: The supply curve will be expected to shift to the left in this situation. For a fully accurately labelled diagram that is clearly micro based and focused on the impact on a single product (1) that is used to clearly explain the resulting change in the price depending on whether demand is price elastic or price inelastic (1) and quantity supplied (depending on whether supply is price elastic or price inelastic/use of buffer stocks/ effect of improved productivity) (1)	8

Question	Answer	Marks
1	AO3 Evaluation (max 2 marks) EITHER: The extent of the change will be determined by whether the good is a normal good or an inferior good / the value of p_{ed} / the value of Y_{ED} / the value of PES / the size of the actual change in income. A discussion focused on the extent of change (1) followed by a justified conclusion (1) OR: The extent of the change will depend on the importance of wage costs to an individual firm / the significance of PES (1) A discussion focused on the extent of change followed by a justified conclusion (1)	
	AO1 Knowledge and understanding	3
	AO2 Analysis	3
	AO3 Evaluation	2

Question	Answer	Marks
2	With the help of a demand and supply diagram, explain how the introduction of an indirect tax affects equilibrium in a market <u>and</u> consider the extent to which the incidence of the tax will fall on the consumer. Follow the point-based marking guidance at the top of this mark scheme and award: up to 3 marks for AO1 Knowledge and understanding up to 3 marks for AO2 Analysis up to 2 marks for AO3 Evaluation. AO1 Knowledge and Understanding (max 3 marks) An accurate definition of an indirect tax (1) shown through an accurate diagram, correctly labelled axes and equilibrium price and quantity (1). Understanding that indirect taxes are used to discourage the consumption of demerit goods (1). AO2 Analysis (max 3 marks) Uses the diagram to explain that an indirect tax shifts the supply curve to the left (1), which causes an increase in the price of the good to the consumer (1) and reduces the quantity demanded (1). AO3 Evaluation (max 2 marks) One mark for a valid explanation that the incidence of tax borne by the consumer will depend on the PED of the product (1), (and that the more price inelastic demand is the greater the proportion of the tax is passed onto the consumer (1)), with a second mark for a justified conclusion.	8
	AO1 Knowledge and understanding	3
	AO2 Analysis	3
	AO3 Evaluation	2

Question	Answer	Marks
3	Explain the functions of price in resource allocation and consider the importance of these functions in relation to the potential effectiveness of a market economy. Follow the point-based marking guidance at the top of this mark scheme and award: - up to 3 marks for AO1 Knowledge and understanding up to 3 marks for AO2 Analysis - up to 2 marks for AO3 Evaluation. AO1 Knowledge and understanding (max 3 marks) A clear knowledge and understanding of at least two functions of price in resource allocation: - rationing (1) - signalling (transmission of preferences) (1) - incentivising (1) Two functions well explained can gain 3 marks; 1 function explained can achieve a maximum of 1 mark. AO2 Analysis (max 3 marks) An analysis of the importance of rationing (1), signalling (transmission of preferences) (1) and incentivising (1) to the operation of a market economy. Note: this must be in the context of a market economy. AO3 Evaluation (max 2 marks) Offers a valid judgement on the importance of at least two functions in relation to the potential effectiveness of a market economy (1) to reach a justified conclusion. (1)	8
	AO1 Knowledge and understanding	3
	AO2 Analysis	3
	AO3 Evaluation	2

SECTION B

1.	B	11.	B	21.	C
2.	B	12.	A	22.	D
3.	C	13.	D	23.	A
4.	C	14.	B	24.	A
5.	B	15.	A	25.	D
6.	B	16.	D	26.	D
7.	B	17.	C	27.	C
8.	B	18.	C	28.	B
9.	D	19.	B	29.	C
10.	C	20.	A	30.	C